ATC System — CRC Cards

CSCI 363 — Assignment 4

TransponderPacket

Responsibilities	Collaborators
• Represent a raw high-density broadcast packet received from an aircraft • Store encoded flight data fields (aircraft ID, type, position, altitude, speed, heading) • Expose raw byte payload for decoding	• PacketDecoder

GroundReceiver

Responsibilities	Collaborators
• Listen continuously on the radio frequency for incoming transponder broadcasts • Wrap each received signal into a TransponderPacket • Forward each packet to the PacketDecoder • Log reception timestamps	• TransponderPacket • PacketDecoder

PacketDecoder

Responsibilities	Collaborators
• Receive a raw TransponderPacket from the GroundReceiver • Parse and decode the high-density packet format into structured fields • Produce a validated FlightRecord for storage • Reject or flag malformed packets	• TransponderPacket • FlightRecord • GroundReceiver

FlightRecord

Responsibilities	Collaborators
• Hold current decoded state for one aircraft (ID, type, position, altitude, speed, heading, timestamp) • Update fields when newer data arrives • Provide a snapshot of flight state to querying classes • Track last-seen time to support stale-data detection	• AircraftDatabase • PacketDecoder

AircraftDatabase

Responsibilities	Collaborators
• Store and index all active FlightRecords keyed by aircraft ID • Insert a new FlightRecord when an unknown aircraft is first seen • Update an existing FlightRecord when new data arrives • Remove or archive records for aircraft no longer transmitting • Return a FlightRecord by aircraft ID on query • Return the full list of current FlightRecords for display and analysis	• FlightRecord • DisplayRefreshScheduler • DangerAnalyzer • ControllerQueryHandler

DisplayRefreshScheduler

Responsibilities	Collaborators
• Trigger a display refresh cycle every 10 seconds • Fetch the latest list of FlightRecords from AircraftDatabase • Push updated flight data to RadarDisplay for rendering	• AircraftDatabase • RadarDisplay

RadarDisplay

Responsibilities	Collaborators
• Render aircraft positions as icons on the radar screen • Display flight labels (ID, altitude, speed) beside each aircraft icon • Highlight aircraft with active DangerAlerts visually • Refresh the full screen when instructed by DisplayRefreshScheduler • Forward controller click/query events to ControllerQueryHandler	• FlightRecord • DangerAlert • DisplayRefreshScheduler • ControllerQueryHandler

DangerAnalyzer

Responsibilities	Collaborators
• Retrieve all current FlightRecords from AircraftDatabase • Compute projected positions using each aircraft's speed and heading • Detect proximity violations between any pair of aircraft • Detect altitude conflicts within a danger threshold • Detect aircraft deviating from expected flight parameters • Produce a DangerAlert for each detected dangerous situation	• AircraftDatabase • FlightRecord • DangerAlert • AlertNotifier

• Forward active DangerAlerts to AlertNotifier	

DangerAlert

Responsibilities	Collaborators
• Represent a single detected dangerous situation • Record the involved aircraft IDs, alert type, severity level, and timestamp • Provide a human-readable description of the danger for display	• DangerAnalyzer • AlertNotifier • RadarDisplay

AlertNotifier

Responsibilities	Collaborators
• Receive DangerAlerts from DangerAnalyzer • Trigger audible and visual alarm signals for the controller • Log each alert with timestamp to the alert history • Suppress duplicate alerts already acknowledged by the controller	• DangerAlert • RadarDisplay

ControllerQueryHandler	
Responsibilities	**Collaborators**
• Accept a query request from the controller (by aircraft ID or screen selection) • Retrieve the matching FlightRecord from AircraftDatabase • Retrieve any active DangerAlerts for the queried aircraft • Format and return a detail view to the RadarDisplay	• AircraftDatabase • FlightRecord • DangerAlert • RadarDisplay